

# Week 8 teaching guide

Test and improve · GROW computing · 40 minutes

**I can test my game against clear expectations and explain one improvement.**

AQA 71638 Programming with Scratch unit 6

## Before the lesson

Open your own W07\_Challenge.sb3 and the bug project. Have the test record, the named booklet and the approved save location ready.

Teacher example: W08\_Teacher\_Model.sb3. Pupil access file: W08\_Recovery\_Challenge.sb3. Recovery supplies prior work only and does not evidence authorship.

Independent Task: allow 25-32 for making and testing, then 32-35 for an individual demonstration and pupil voice.

| Stage and time           | Teach and check                                                                             |
|--------------------------|---------------------------------------------------------------------------------------------|
| Arrival / 0-3            | Open your own W07_Challenge.sb3. Put Player at the safe start.                              |
| Starter / 3-6            | "It is broken" does not help. "The down arrow moves Player up" does help.                   |
| I Do / 6-11              | Use the named setup for each test. Repeat it when retesting. Write what you expect first.   |
| We Do / 11-18            | Arrows. Walls. Finish. Feature. Restart. Run the five checks on the shared game.            |
| I Do 2 / 18-21           | Change one cause only. Run the same test again.                                             |
| We Do 2 · Task / 21-25   | Choose a peer or teacher tester. Agree how the feedback is given.                           |
| Independent Task / 25-35 | Run the five checks on your own game. Make one useful improvement.                          |
| Exit / 35-40             | Save as W08_Final.sb3. Show your test record and a repair, improvement or agreed next step. |

## Choose the task route

Step by step: Use the five-check record. Say or point to each result.

Main task: Run all five checks. Record results and retest any change.

Challenge: After the core checks, investigate a corner or contact during a glide.

# Teaching notes and answers

Use the teacher model for demonstration, then return to pupil construction

“It works” is not a test result. A test names the start, the expected result and the actual result. A repair is only finished once the same check is run again.

## Block and design reference

| Find this      | Use it like this                                                                                       |
|----------------|--------------------------------------------------------------------------------------------------------|
| Same start     | Use a known setup for each check. Repeat that setup when retesting. Test restart after a win.          |
| Expected first | Write the expected result before running, so the run tests the prediction and not the memory.          |
| One cause      | Repair one cause at a time. Two changes at once hide which one worked.                                 |
| Retest         | Retest the same check, then the earlier checks. A repair that breaks an earlier check is not finished. |
| The record     | Test notes support the teacher record. The AQA summary sheet stays the official evidence.              |

## Expected answers

1. “It is broken” gives no test and no result. A useful report names the check, the start, the expected result and the actual result.
2. The bug project has three faults. Down arrow uses change y by 10, so Player moves up. The wall check has no forever, so it runs once at the flag and never again. The win script has no empty say, so the old message survives a restart.
3. The arrows check finds the first, the walls check finds the second and the restart check finds the third. The finish and feature checks pass. Passing checks are results too: they show what the repair must not break.
4. Change one cause, then run the same check again. Then run the four other checks for any new faults.
5. An edge case starts from a known place and tries the awkward thing: a corner, a narrow gap, or contact during a glide. An ordinary run misses it because an ordinary run avoids it.
6. Outcome 6 is testing, observed. Review outcomes 1 to 5 individually now. A partly finished game gets a recorded next step, not an assumed completion.

## A likely misconception

“It works” is not a test result. A test names the start, the expected result and the actual result. A repair is only finished once the same check is run again.

# Access and evidence

Record the pupil action and the support used

## Flexible support

| Current need               | Adaptation                                                                                                 |
|----------------------------|------------------------------------------------------------------------------------------------------------|
| Reading or memory          | Use short instructions, the reference page and one action at a time.                                       |
| Communication              | Accept speech, pointing, drawing or exact-word scribing.                                                   |
| Keyboard or pointer access | Use an agreed school access device, or let the pupil direct an adult and record who decided and who acted. |
| Regulation or attention    | Offer quiet work, a pause and the same return step.                                                        |
| Ready to transfer          | Use Challenge after the common goal. Ask for a reason with a test result.                                  |

## Pupil voice with a response

What did your tester's feedback change? Name the actual improvement, or explain why a suggestion was deferred.

Ask "What I said, and what it changed". Listen and reply: use it now, adapt it together, or defer it with a reason and a time.

## Individual evidence

AQA 71638 outcome 6: Observe the pupil testing the game, recording expected and actual results, and retesting after a repair. Review all six outcomes individually before any award claim.

| Pupil action or check                                                        | Date and support used |
|------------------------------------------------------------------------------|-----------------------|
| I ran all five checks using the named setups.                                |                       |
| I recorded expected and actual results, not only 'it works'.                 |                       |
| I recorded a fault and repair, or an all-pass result and useful improvement. |                       |
| I retested the same steps and the earlier checks still pass.                 |                       |

Record completed outcomes on the AQA summary sheet. Do not infer creation from supplied code.

## If a device is unavailable

Use the five named checks to rehearse expected results. Record actual results only after testing on a device. Complete any repair and retest there.

# Shared checks and teaching notes

Week 8 Classroom reference

| Check     | Setup and expected behaviour                                                                                                                      |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Arrows  | Flag before each key. Right / left / up / down. Each key moves Player in its named direction.                                                     |
| 2 Walls   | Flag. Try clear space, then two different walls. Clear space: continue. Each wall: return to the safe start.                                      |
| 3 Finish  | Flag away from Finish. Reach Finish, then stay. No early win. One win response when Finish is reached.                                            |
| 4 Feature | Flag. Watch a full outward and return journey. Avoid Hazard, then touch it. Hazard repeats. Avoidance leaves Player free. Contact returns Player. |
| 5 Restart | Win first, then flag. Try a key, a wall and a new win. Old message clears. Player and Hazard reset. The game works again.                         |

The shared bug file contains three deliberate faults: down moves up, the wall check runs only at the flag, and restart leaves the old message. Finish and Hazard checks should pass. The HTML model tests movement and contact, not only appearance.

Use the HTML probes for the shared diagnosis within seven minutes. In Scratch, a teacher may position Player at the named test location after the flag to isolate a rule. Pupils must also complete an ordinary playable route. Never position a sprite and call that evidence of creating its code.

An all-pass result is valid. AQA outcome 6 asks for testing; finding a fault and writing test notes are local teaching activities. Repair if needed, or improve the playing instructions and retest. Record unfinished work honestly and agree time to finish it.

AQA summary sheets remain the official evidence. Use the final evidence check to locate individual work for outcomes 1 to 6. A saved teacher model cannot establish pupil authorship.